

PROJECT TITLE:

**EXPLORATORY DATA ANALYSIS OF HEART STROKE
PREDICTION DATASET**

Work submitted by:

Medicine group 2

As partial fulfilment of the requirements for the

TechRise Programme (AI/ML Track)

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1. Introduction

A stroke occurs when the blood supply to a part of the brain is occluded, thus preventing brain tissues from getting oxygen and nutrients. Within minutes brain cells begin to die, making it a medical emergency that requires immediate intervention. Clinically, strokes are categorized into 2: **Ischemic and hemorrhagic stroke**. Ischemic stroke is caused due to occlusion of blood vessel and hemorrhagic stroke is caused due to rupture of blood vessel.

WHO Perspective

According to the World Health Organization (WHO), cardiovascular diseases (CVDs), including strokes, remain the leading cause of death globally, claiming an estimated 17.9 million lives each year. The WHO highlights that over three quarters of CVD deaths take place in low and middle income countries making stroke not just a clinical issue, but a critical global socioeconomic challenge.

Rationale for dataset selection

To investigate this critical health issue in alignment with global health priorities, the Heart Stroke Prediction dataset was specifically chosen because it offers a holistic, multi-dimensional view of patient health profiles.

It tracks essential features across 12 key attributes:

Demographics: gender, age, ever_married, work type and residence type

Clinical indicators: hypertension, heart disease, avg glucose level, and bmi

Behavioral factors: smoking status

Objective of the analysis

Identify which demographic and clinical variables share the strongest statistical correlation with stroke events.

Data set Description

Item	Description
Dataset name	Heart_Stroke Dataset
Source	Kaggle
Number of observations	5110
Number of features	10
Target Variable	Stroke
Missing values	bmi(201)
Numerical Variables	Age, hypertension,heart disease,heart disease,avg_glucose_level and bmi.
Categorical Variables	Gender,ever_married,work_type,Residence_type,smoking status

Health_Stroke.head()

	id	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	stroke
0	9046	Male	67.0	0	1	Yes	Private	Urban	228.69	36.6	formerly smoked	1
1	51676	Female	61.0	0	0	Yes	Self-employed	Rural	202.21	NaN	never smoked	1
2	31112	Male	80.0	0	1	Yes	Private	Rural	105.92	32.5	never smoked	1
3	60182	Female	49.0	0	0	Yes	Private	Urban	171.23	34.4	smokes	1
4	1665	Female	79.0	1	0	Yes	Self-employed	Rural	174.12	24.0	never smoked	1

Health_stroke.info()

```
Health_stroke.info()
✓ 0.0s

<class 'pandas.DataFrame'>
RangeIndex: 5110 entries, 0 to 5109
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   id                    5110 non-null   int64
1   gender                5110 non-null   str
2   age                   5110 non-null   float64
3   hypertension          5110 non-null   int64
4   heart_disease         5110 non-null   int64
5   ever_married          5110 non-null   str
6   work_type              5110 non-null   str
7   Residence_type        5110 non-null   str
8   avg_glucose_level     5110 non-null   float64
9   bmi                   4909 non-null   float64
10  smoking_status        5110 non-null   str
11  stroke                5110 non-null   int64
dtypes: float64(3), int64(4), str(5)
memory usage: 479.2 KB
```

Health_Stroke.describe()

```
Health_stroke["age"].describe()
```

✓ 0.0s

```
count    5110.000000
mean      43.226614
std       22.612647
min        0.080000
25%       25.000000
50%       45.000000
75%       61.000000
max       82.000000
Name: age, dtype: float64
```

```
Health_stroke["avg_glucose_level"].describe()
```

✓ 0.0s

```
count    5110.000000
mean     106.147677
std       45.283560
min       55.120000
25%       77.245000
50%       91.885000
75%      114.090000
max      271.740000
```

Health_Stroke.isnull().sum()

```
#Missing Values
```

```
Health_stroke.isnull().sum()
```

✓ 0.0s

```
id          0
gender      0
age         0
hypertension 0
heart_disease 0
ever_married 0
work_type   0
Residence_type 0
avg_glucose_level 0
bmi        201
smoking_status 0
stroke      0
dtype: int64
```

Exploratory data analysis(EDA)

Descriptive statistics interpretation

Patient Data Intepretation Report

1. Age Profile

Average Patient Age	43.26 years
Median Age (50th Percentile)	Half of the patient are older than 45 Years
Mininum Age	0.08 years
Maximum Age	82 years
25th percentile	25% of the patients are 25 years old or younger.
75 th percentile	75% of the patients are 61 years old or younger.

2. Average glucose levels

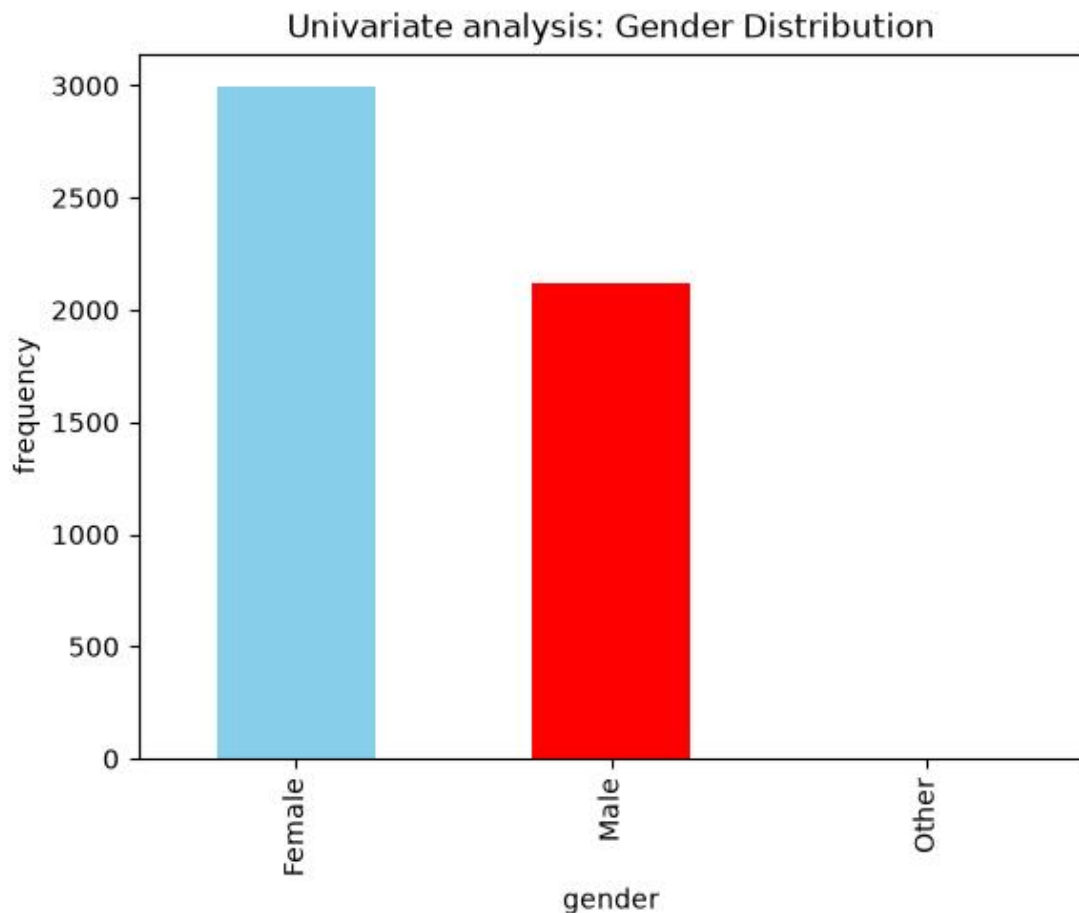
Average glucose level	106mg/dl
Median glucose level	Half of the patients have an average glucose level Of 91.89mg/dl
Minimum recorded	55.12 mg/dl
Maximum recorded	271.74mg/dl

3. Body Mass Index (BMI)

Average BMI	28.8
Median BMI	Half of the patients have a BMI of 28.10
Minimum recorded	10.30
Maximum recorded	97.60

Univariate Analysis

On gender distribution



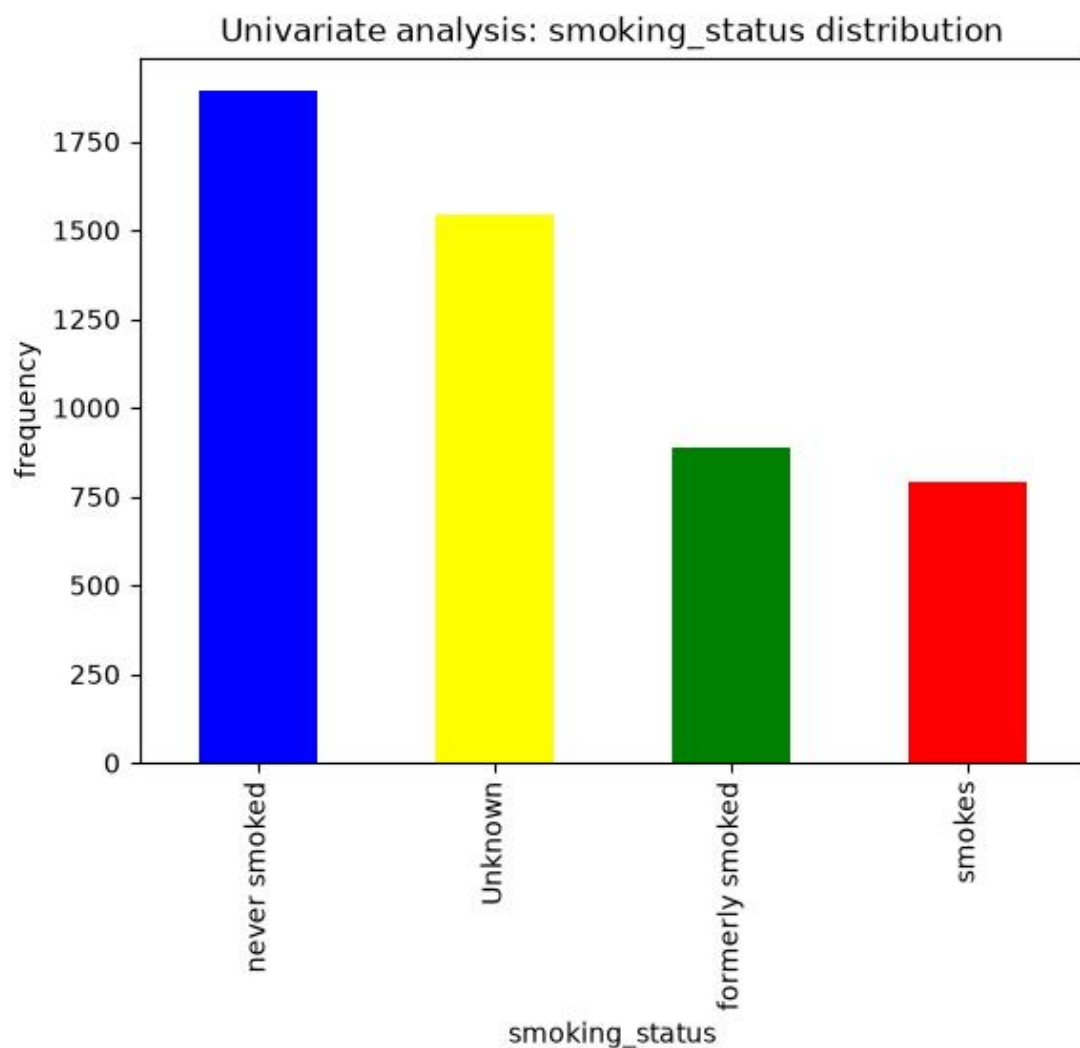
Observations

1. Females represent the largest group in the dataset with a frequency of almost 3,000
2. Males make up the second largest group with a frequency of about 2100
3. The “Other” category has an extremely low frequency.

Interpretations

The dataset shows gender distribution with a higher representation of female individuals compared to male individuals.

Smoking status distribution



Observations

1. Individuals who have never smoked have the largest frequency in the dataset.
2. The unknown categories is the second largest capturing exactly 1500 entries.

3. The formerly smoked individuals has a frequency of roughly 850.

4. Individuals who currently smokes form the smallest group at approximately 750.

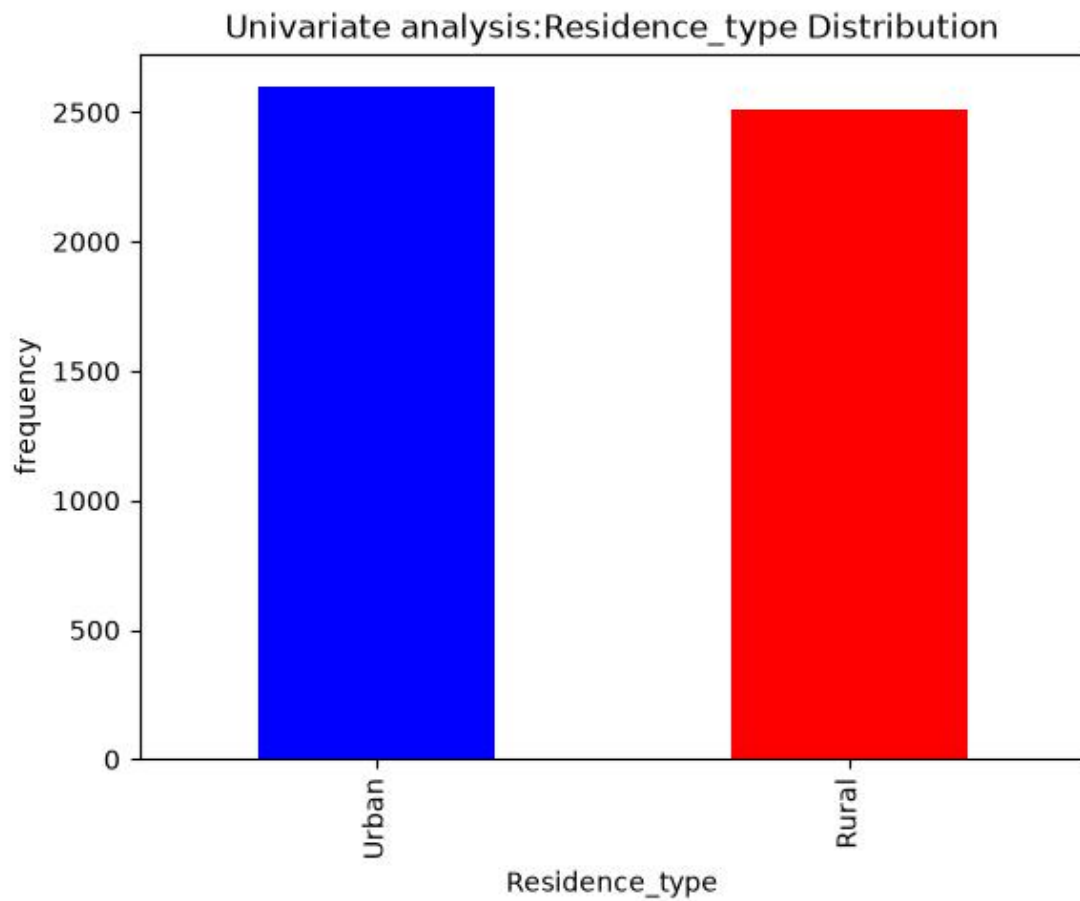
Interpretation

1. The majority of the sampled population consists of non smokers, outnumbering active and former smokers combined.

2. The notably high frequency of the unknown category indicates a substantial amount of missing or unrecorded behavioural data.

3. Current and former smokers represent minority segments of the dataset. This distribution suggests that if this feature is used to predict a health outcome, the model will have significantly more exposure to non smoking histories than active smoking behaviors.

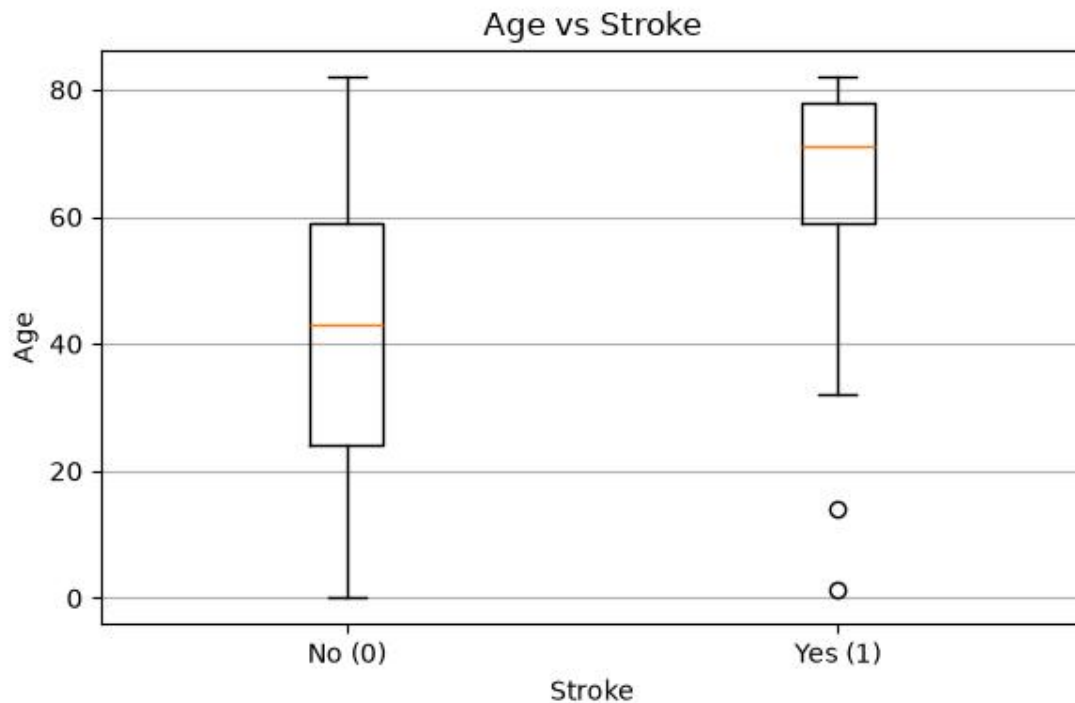
Residence type distribution



Observations

1. From the dataset, most individuals live in the urban area than in the rural areas.

Bivariate analysis



Observations

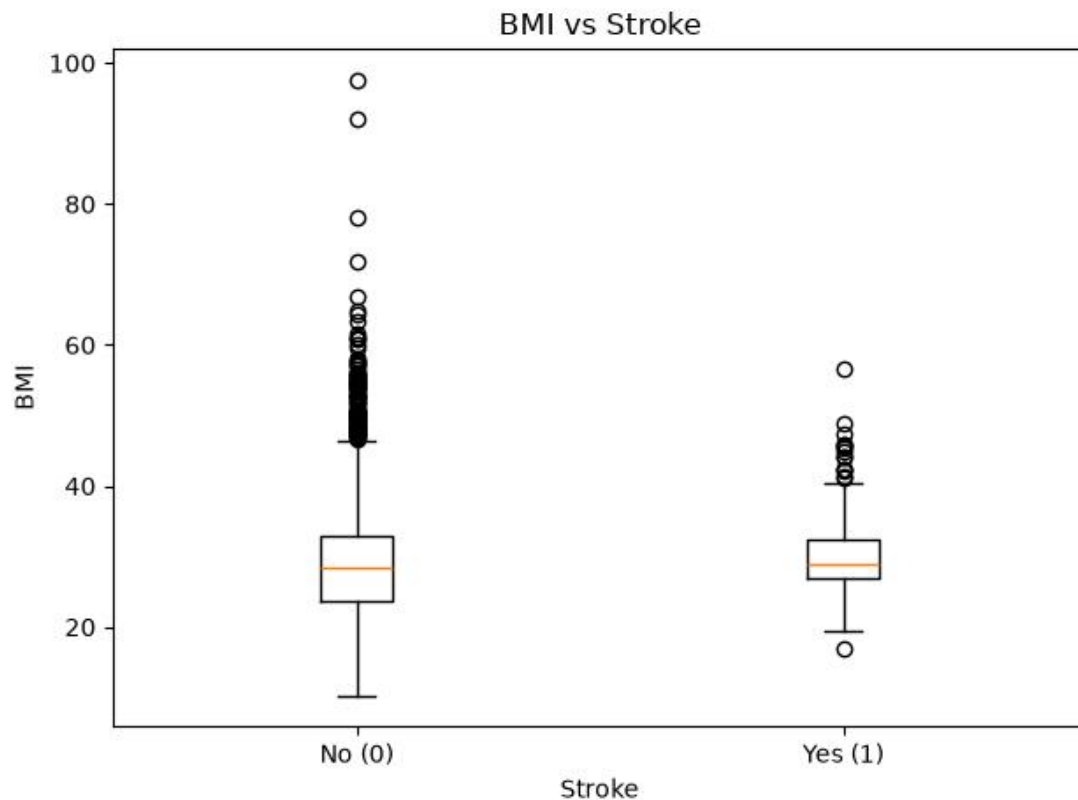
1. Significance difference in median age
No Stroke(0): The median age(represented by the orange line inside the box) is roughly 43 years old.

Stroke (1): The median age is higher, sitting at around 70 years old.

Interpretations

This indicates that individuals who experience a stroke in this dataset tend to be significantly older than those who do not.

Bmi vs stroke



Obseervations

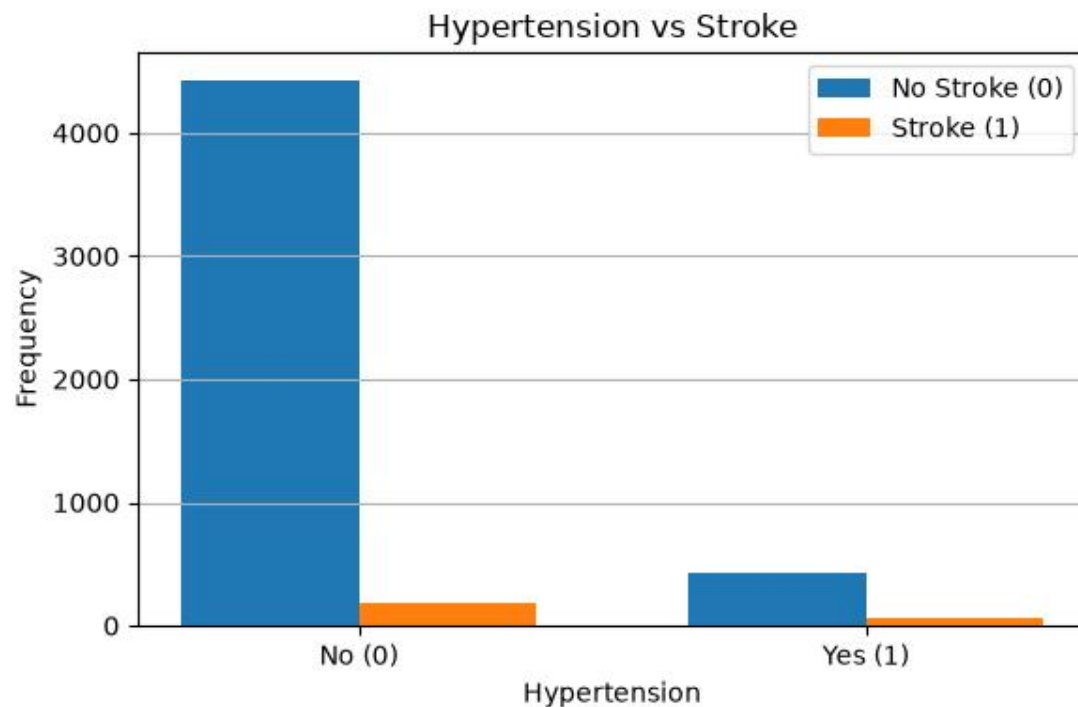
Similar median bmi

The orange median lines inside the boxes are sitting at almost the exact the same level.

Interpretations

The median BMI for individuals who did not have a stroke and those who did have the stroke is virtually identical, hovering right around 28 to 29.

Hypertension vs stroke analysis



Observations

1.Imbalance in the dataset

The blue bars are taller than the orange bars across both the "No(0)" and "Yes(1)" hypertension categories.

Interpretations

The data-set is highly imbalanced, meaning that vast majority of individuals in this study did not experience a stroke.

Most individuals in this data set do not have high blood pressure. While the absolute number of strokes is low in both categories due to the class imbalance, the proportion of people who suffer stroke is noticeably higher among individuals who have hypertension.

Summary

Hypertension shows a clear distribution with stroke status. From the data set, if a patient has hypertension, their visual likelihood of being in the stroke category increases compared to those without high blood pressure.

Story Telling

The analysis indicates that stroke occurrences are heavily concentrated among older individuals and are strongly associated with hypertension. While the majority of the patients do not have high blood pressure, the proportion of individuals who suffer a stroke is visibly and significantly higher among the hypertensive population.

This finding is highly critical because it highlights a compounding clinical risk. As the body ages, blood vessels lose their elasticity, and when this physiological aging is combined with the constant mechanical strain of hypertension, the risk of vascular event increases exponentially.

According to the World Health Organization (WHO), hypertension remains a leading preventable cause of cardiovascular diseases globally, making it a vital diagnostic feature alongside age.

Other physical health indicators in this dataset such as BMI ,showed nearly identical median distributions between stroke and non-stroke patients,proving that typical weight metrics alone are not strong standalone predictors compared to the critical duo of age and blood pressure.

Based on these implications, we recommend establishing regular health screenings focusing on patients over the age of 60,alongside aggressive blood pressure management programs to catch and control hypertension early.

Conclusions

This exploratory data analysis worked through 5,110 patient records containing 12 variables spanning demographic information (age, gender), lifestyle and social factors (work type, residence type, marital status, smoking status), and clinical indicators (hypertension, heart disease, BMI, average glucose level), with stroke occurrence as the target outcome. The workflow followed a structured EDA process: first checking data quality (no duplicate records were found), then assessing missing values (only BMI had gaps, with 201 missing values imputed using the mean), followed by univariate analysis of each variable individually, and finally bivariate analysis pairing each variable against the stroke target using

the appropriate chart type — scatter plots for numerical-vs-numerical relationships, boxplots for numerical-vs-categorical comparisons, and grouped bar charts for categorical-vs-categorical comparisons. It's worth noting upfront that the dataset is heavily imbalanced, with only 249 patients (4.9%) having experienced a stroke compared to 4,861 (95.1%) who had not — this imbalance doesn't invalidate the patterns found, but it does mean any future predictive model built on this data will need to account for it (e.g., through resampling or class-weighting), since a model could otherwise achieve high accuracy just by predicting "no stroke" every time.

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